REDWAN
Ahamad Samir

AI Researcher & Aspiring RL Engineer

BSc CSE | ULAB | Expected 2026

Contact

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Google Scholar
0009-0008-3446-0845
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Portfolio
Nurbag, Citashal, Pagla, Fatullah, Narayangonj

Technical Skills

Languages: C, C++, C#, Python, Java, Dart, JavaScript, SQL

Frameworks: Django, Flutter, OpenCV, TensorFlow, Keras, PyTorch

Web: HTML, CSS, JavaScript, REST APIs, Firebase

ML/DL: CNNs, RNNs, Transfer Learning, RAG, LLMs, NLP

Research: LLMs, AGI, RL, Adaptive Learning, Edge AI

Systems: Signal Processing, Embedded, IoT, Multi-Agent

Certifications

- BTEB (Computer Office Application)
- Creative IT (Ethical Hacking)
- Enayetpuri (Microsoft Office)

Languages

English (Fluent)
Bangla (Native)
French (Intermediate)

Personal Details

DOB: 20 Aug 2003
Nationality: Bangladeshi
Religion: Islam
Blood: A+

Reference

Muhammad Golam Kibria, PhD, SMIEEE
Professor & Head | CSE
Acting Head | EEE
University of Liberal Arts Bangladesh
+88-02-223328001-6
golam.kibria@ulab.edu.bd

Professional Summary

CS undergraduate with 3 peer-reviewed publications in IEEE and international conferences. Specializes in deep learning, computer vision, and LLM systems. Experienced in production-ready AI from healthcare diagnostics to smart city infrastructure. Researching adaptive RAG and multi-agent RL for AGI.

Education

Bachelor of Science in Computer Science and Engineering
University of Liberal Arts Bangladesh **Expected 2026**
GPA: 3.46/4.0 | Specialization: AI & Machine Learning
Coursework: Deep Learning, Computer Vision, NLP, ML Algorithms, Data Mining, Software Engineering, Signal Processing

Research Publications

[1] **HornNet: Two-Tier DL for Vehicle Classification from Acoustic Horn Signatures** IEEE QPAIN 2026
Real-time audio-based vehicle classification with hierarchical edge-detection + deep classifier. Privacy-preserving smart city monitoring. [DOI]

[2] **Early-Stage Coronary Artery Disease Prediction from Coronary Angiogram** BMI 2025
CNN with attention for automated stenosis detection in low-resource clinical settings.

[3] **Scalable sEMG-Based User-Independent DL for Wheelchair Control** BMI 2025
Real-time, calibration-free assistive navigation using multi-channel surface EMG signals.

Professional Experience

Research Intern May 2024 – Present
Tiny Neuron Research Group
- Designed NN architectures for medical imaging and signal processing
- Led 3 peer-reviewed research projects to IEEE/international publication
- Developed edge-AI deployment pipelines for resource-constrained environments

Head of Technical & Web Development May 2024 – Mar 2025
PR4U
- Led full-stack development, deployment, and maintenance for PR campaigns
- Integrated AI analytics improving performance and engagement by 40%
- Managed cybersecurity and infrastructure for 500+ active users

Graphic Designer Jan 2022 – Jun 2022
Enayetpuri Computer Training Academy
- Designed marketing collateral using Adobe InDesign

Key Projects

1. Completed Projects
- **BiyerBajar (বিয়ের বাজার):** Flutter wedding marketplace with Bangla budget calculator, blind reverse-bidding, and glassmorphic UI.
- **Facial Attendance System:** OpenCV/Python facial recognition with real-time logging.
- **Lost And Found:** Full-stack Django app for lost item recovery with location search.
- **Robotic Arm:** ESP-NOW wireless robotics with sensor integration.
- **Smart Home App:** Flutter IoT controller via Firebase and MQTT.
- **Justice Link BD:** Flutter corruption reporting with CNN fake detection, SOS geo-fencing, and case tracking.
- **Tiny Neuron Research Web:** Company website for research group.

- **AI Talent Matching:** Platform reducing hiring workload by 90% via AI matching and auto-interviews.
- **Medical Chatbot:** Symptom analyzer with rule-based AI and LLM backend with conversational memory.

2. Ongoing Projects

- **Lexical AI:** Browser-native AI speech therapist using Web Audio API + RL for personalized vocal rehabilitation.
- **Next Gen RAG:** Self-improving RAG for LLMs with adaptive learning from user feedback.

3. Upcoming Projects

- **Smart City Twin (MARL):** Multi-agent RL for urban optimization using OSM and SUMO.
- **Smart Transport:** Flutter app with digital bus card and in-app recharge.
- **Car Health Predictor:** Flutter + IoT device with DL failure prediction and maintenance alerts.

Career Objective

To advance AI through research in adaptive learning, multi-agent systems, and AGI. Seeking roles bridging cutting-edge research with real-world deployment for measurable social impact.